

Program : Diploma in Textile Technology	
Course Code : 4068	Course Title: Fabric Manufacture II Lab
Semester : 4	Credits: 1.5
Course Category: Program core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide thorough knowledge on setting the various primary, secondary and auxiliary mechanisms in power loom.
- To impart knowledge of setting dobby in power loom.
- Analyzing different sample cloths of compound structures and find the requirement to produce the design in a loom.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics 1	1
Basic Knowledge of textile mechanics lab		Basic textile mechanics lab	2
Basic Knowledge of fabric manufacturing		Fabric manufacture I	3

Course Outcomes:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the setting of primary and secondary mechanisms in power loom according to its timing.	8	Applying
CO2	Explain the settings of various auxiliary motions in power loom.	10	Applying
CO3	To impart knowledge of timing and setting of various dobbies.	10	Applying
CO4	Analyzing different sample cloths of compound structures and find the loom requirement to produce the design.	14	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3		2			
CO2	3	3		2			
CO3	3	3		2			
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the setting of primary and secondary mechanisms in power loom according to its timing.		
M1.01	Study the mechanism, dismantle and assemble the parts of plain tappet shedding mechanism.	1	Applying
M1.02	Study the mechanism, dismantle and assemble the parts of cone over picking mechanism	1	Applying
M1.03	Study the mechanism of beat up and verify eccentricity of sley	2	Applying
M1.04	Study the mechanism, dismantle and assemble the parts of negative let off motion	2	Applying
M1.05	Study the mechanism, dismantle and assemble the parts of seven wheel take up motion also calculate the dividend and determine the change wheel when standard wheel changes.	2	Applying
CO2	Explain the settings of various auxiliary motions in power loom.		
M2.01	Study the mechanism, dismantle and assemble the parts of side weft fork motion	2	Applying
M2.02	Study the mechanism, dismantle and assemble the parts of warp stop motion.	2	Applying
M2.03	Study the mechanism, dismantle and assemble the parts of loose and fast reed motion	3	Applying
M2.04	Study the mechanism, dismantle and assemble the parts of check strap and temples	3	Applying
	Lab Exam- I	1.5	

CO3	To impart knowledge of timing and setting of various dobbies.		
M3.01	Study the mechanism, dismantle and assemble the parts of climax dobby.	2	Applying
M3.02	Prepare the Design, draft, Peg plan and peg the lags suitable for RH & LH Dobby	2	Applying
M3.03	Study the mechanism, dismantle and assemble the parts of Knowles dobby.	3	Applying
M3.04	Prepare the Design and arrange the rollers and bushes according to the design for the working of Knowles dobby.	3	Applying
CO4	Analyzing different sample cloths of compound structures and find the loom requirement to produce the design.		
M4.01	Analyze the Ends and Picks per unit length, Count of Warp and Weft, Identify Material of Warp and Weft, Determine Crimp of Warp and Weft, Draw Design, Draft, Lifting – plan & denting plan for compound structures.	4	Applying
M4.02	Analyze the Ends and Picks per unit length, Count of Warp and Weft, Identify Material of Warp and Weft, Determine Crimp of Warp and Weft, Draw Design, Draft, Lifting – plan & denting plan for Extra Warp and Extra Weft fabric	4	Applying
M4.03	Analyze the Ends and Picks per unit length, Count of Warp and Weft, Identify Material of Warp and Weft, Determine Crimp of Warp and Weft, Draw Design, Draft, Lifting – plan & denting plan for Net and Leno Fabrics.	4	Applying
M4.04	Estimate the quantity of warp and weft required for a given piece of cloth (plain and striped) of 100 meters	2	Applying
	Lab Exam - II	1.5	

The students should keep separate records for 1. Fabric Manufacture and 2. Cloth samples (minimum 50 samples) and submit certified & bonafide records at the time of practical examination

Text / Reference:

T/R	Author/Book Title
R1	Thomas W.Fox - The Mechanism of Weaving - Universal Publishing corp.,Mumbai
R2	A.Ormerod - Modern Preparation and Weaving Machinery - Woodhead publishing Ltd.,England
R3	R. Sengupta - Weaving Calculations - D. B. Taraporevala sons & co Ltd
R4	NCUTE - Woven Fabric Production II – NCUTE New Delhi
R5	Prof. J. C. Chakravorthy - Mechanism of Weaving Vol- I&II- West Bengal

Online Resources:

Sl.No	Website Link
1	http://www.nptel.ac.in/courses
2	http://freetexbooks.blogspot.in/2011/10/handbook-of-weaving.html
3	https://www.textileschool.com
4	http://textilelearner.blogspot.in
5	https://textilestudycenter.com